In-Class Activity: Bioavailability

Module 7a - Bioavailability and Bioequivalence · In-class activity · 30 Sep 2026

Name _____ Date _____

1a After oral administration of a single 500 mg tablet of an antihypertensive drug to a normal volunteer, the calculated AUC was 70 mg h/L. Following a 500 mg IV bolus dose of the same drug to the same volunteer, the elimination half-life was 3 hours and the apparent volume of distribution was 25 L. What is the absolute bioavailability of this oral tablet in this volunteer?

1b Following a 500 mg IV bolus dose of an antihypertensive drug to a normal volunteer, the elimination half-life was 3 hours, the apparent volume of distribution was 25 L, and the AUC was 86.58 (mg/L)hr. What is the expected AUC after administration of a single IV bolus dose of 250 mg to the same volunteer?

3a The equation for concentration of drug in the plasma as a function of time following a 500 mg IV bolus dose was found to be $C_p = 50e^{-0.2t}$. What is the clearance of the drug?

3b A 500 mg IV bolus dose of a drug gives $C_p = 50e^{-0.2t}$, and the clearance of the drug is 2 L/hr. What is the AUC of the drug following this 500 mg IV bolus dose?

3c A 500 mg IV bolus dose of a drug gives an AUC of 250 (mg/L)hr. If the AUC following a 500-mg oral tablet dose was found to be 188 mcg hr/mL, what is the oral bioavailability of the drug from the tablet?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a

Her key

Answer: 0.81 (as a decimal fraction)

WHAT THE STEM GIVES

D_{po}	500 mg tablet	the denominator of the dose ratio, which is 1
AUC_{po}	70 mg h/L	the numerator of the area ratio
D_{IV}	500 mg IV bolus	the numerator of the dose ratio, and of $AUC_{IV} = \frac{D_{IV}}{Cl}$
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	25 L	in $Cl = kV_D = 0.231 \times 25 = 5.775$ L/hr

THE WORKING, ONE LINE AT A TIME

UNIT

$$k = \frac{0.693}{3\ hr} = 0.231\ hr^{-1};\ Cl = k \times V_D = (0.231\ hr^{-1})(25\ L) = 5.775\ L/hr$$

No IV AUC is given, so it has to be built from the IV data. The half-life gives k, the elimination rate constant, and k times V_D (the apparent volume of distribution) is the clearance, in litres per hour.

SETUP

$$AUC_{IV} = \frac{D_{IV}}{Cl}$$

$D_{IV} = Cl \times AUC_{IV}$ is on the equation sheet: clearance is the constant that relates a dose to its area. Rearranged, the IV dose over the clearance is the IV AUC.

ALGEBRA

$$AUC_{IV} = \frac{500\ mg}{(25\ L)(0.231\ hr^{-1})} = \frac{500\ mg}{5.775\ L/hr} = 86.58\ (mg/L)hr$$

mg over L/hr is mg·hr/L, the unit of an AUC. This is the area the same 500 mg would give if all of it were in the circulation.

ALGEBRA

$$F = \left(\frac{70}{86.58}\right)\left(\frac{500\ mg}{500\ mg}\right) = 0.8085$$

The doses are equal, so the dose ratio is 1 and F is the AUC ratio alone. The same answer comes from $F = \frac{Cl \times AUC_{po}}{D_{po}} = \frac{(70)(25)(0.231)}{500} = 0.8085$; either route works.

ROUND

$$F = 0.81$$

Her value, about 81 per cent. F is dimensionless because every unit cancelled. Written as 0.81 or 81 per cent.

DOES THE NUMBER MAKE SENSE?

Clearance 5.775 L/hr means 500 mg IV gives an area of $\frac{500}{5.775}$ = 86.58 (mg/L)hr; the oral 70 is below it, so F is below 1 and comfortably above 0.5: 0.81.

1b Her key

Answer: 43.29 (mg/L)hr

WHAT THE STEM GIVES

t½	3 hours	gives k = $\frac{0.693}{3}$ = 0.231 hr^{-1}
V_D	25 L	in Cl = kV_D = 5.775 L/hr
reference dose	500 mg	the dose that gave the known area; half of it is asked
reference AUC	86.58 (mg/L)hr	*not needed for the line; halving it is the shortcut*
D_{IV}	250 mg	the numerator of $\frac{D_{IV}}{Cl}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC_{IV} = \frac{D_{IV}}{Cl}, \text{ with } Cl = k \times V_D \text{ unchanged}$$

Clearance belongs to the drug and the patient, not to the dose, so the same relation holds with the new dose. The area is proportional to the dose when kinetics are linear (first order).

ALGEBRA

$$AUC_{IV} = \frac{250 \text{ mg}}{(25 \text{ L})(0.231 \text{ hr}^{-1})} = \frac{250 \text{ mg}}{5.775 \text{ L/hr}} = 43.29 \text{ (mg/L)hr}$$

Cl (clearance) = k × V_D, with k (the elimination rate constant) = $\frac{0.693}{3 \text{ hr}}$ = 0.231 hr^{-1} and V_D (the volume of distribution) 25 L, so Cl = 5.775 L/hr. The 250 mg dose over this clearance gives the area: mg over L/hr leaves (mg/L)hr.

ROUND

$$AUC_{IV} = 43.29 \text{ (mg/L)hr}$$

Her value, 43.29: half of the 86.58 found for 500 mg, as expected. With linear pharmacokinetics a proportional change in the dose gives a proportional change in the AUC and in the concentrations.

DOES THE NUMBER MAKE SENSE?

Clearance does not change with the dose, so AUC is proportional to dose: half of 500 mg gives half of 86.58, which is 43.29 (mg/L)hr, below 86.58.

3a Her key

Answer: 2 L/hr

WHAT THE STEM GIVES

D_{IV}	500 mg IV bolus dose	the numerator of $V_D = \frac{D_{IV}}{C_0}$
C_0	50 (the coefficient in Cp = 50e^(−0.2t))	the denominator of $V_D = \frac{D_{IV}}{C_0}$
k	0.2 (the exponent in Cp = 50e^(−0.2t))	multiplies V_D: $Cl = kV_D$

THE WORKING, ONE LINE AT A TIME

SETUP

$C_p = C_0e^{-kt}$: C_0 = 50 mg/L and k = 0.2 hr^{-1}

The IV bolus equation is read directly: the number in front is C_0, the concentration at time zero, and the number in the exponent is k, the elimination rate constant.

ALGEBRA

$$V_D = \frac{D_{IV}}{C_0} = \frac{500\text{ mg}}{50\text{ mg/L}} = 10\text{ L}$$

The apparent volume of distribution comes from the dose over C_0: mg over mg/L leaves litres. The whole dose is in the body at time zero after a bolus, so it is the only moment the amount is known exactly.

ALGEBRA

$Cl = k \times V_D = (0.2\ hr^{-1})(10\text{ L}) = 2$ L/hr

Clearance is the rate constant times the volume: hr^{-1} times L gives L/hr. This clearance is what relates the dose to its AUC in the next part.

ROUND

Cl = 2 L/hr

Cl = 2 L/hr follows from the two readings taken from the equation: V_D (the volume of distribution) = $\frac{500\text{ mg}}{50\text{ mg/L}}$ = 10 L and k = 0.2 hr^{-1}, so Cl = 0.2 × 10 = 2 L/hr. hr^{-1} × L gives L/hr, the volume of plasma cleared of drug each hour.

> **DOES THE NUMBER MAKE SENSE?**
> The 500 mg dose read against the 50 mg/L intercept gives V_D = 10 L, and a fifth of that volume is cleared each hour: 0.2 × 10 = 2 L/hr, above zero.

3b Her key

Answer: 250 (mg/L)hr

WHAT THE STEM GIVES

D_{IV}	500 mg IV bolus dose	the numerator
equation	Cp = 50e^(−0.2t)	*not needed: clearance is given, so C_0 and k stay unused*
Cl	2 L/hr	the denominator

THE WORKING, ONE LINE AT A TIME

SETUP

$$D_{IV} = Cl \times AUC_{IV}, \text{ so } AUC_{IV} = \frac{D_{IV}}{Cl}$$

Clearance is the link between a dose and the area it produces: the dose is the clearance times the AUC (area under the curve). Rearranging gives the area from the dose and the clearance, with no plasma data and no trapezoids.

ALGEBRA

$$AUC_{IV} = \frac{500 \text{ mg}}{2 \text{ L/hr}} = 250 \text{ (mg/L)hr}$$

mg over L/hr leaves mg·hr/L, which is the unit of an AUC. The same figure follows from C_0 over k, 50 over 0.2, for a one-compartment bolus.

ROUND

$$AUC_{IV} = 250 \text{ (mg/L)hr}$$

AUC_{IV} = 250 (mg/L)hr: 500 mg over 2 L/hr, with mg over L/hr leaving (mg/L)hr. This IV area is the reference for the absolute bioavailability of an oral dose of the same drug, $F_{abs} = (\frac{AUC_{po}}{AUC_{IV}})(\frac{D_{IV}}{D_{po}})$, where it sits in the denominator because the IV dose puts the whole dose into the circulation.

> **DOES THE NUMBER MAKE SENSE?**
>
> AUC is also $\frac{C_0}{k} = \frac{50}{0.2} = 250$ (mg/L)hr for a one-compartment bolus, so the two routes agree: 500 mg over 2 L/hr is 250.

3c

Her key

Answer: 0.752 (as a decimal fraction)

WHAT THE STEM GIVES

D_{IV}	500 mg IV bolus dose	the numerator of the dose ratio, which is 1
AUC_{IV}	250 (mg/L)hr	the denominator of the area ratio
D_{po}	500-mg oral tablet dose	the denominator of the dose ratio, which is 1
AUC_{po}	188 mcg hr/mL	the numerator; mcg hr/mL is the same unit as (mg/L)hr

THE WORKING, ONE LINE AT A TIME

SETUP

$$F_{abs} = (\frac{AUC_{po}}{AUC_{IV}})(\frac{D_{IV}}{D_{po}})$$

The tablet is compared with the IV bolus of the same drug, so this is an absolute bioavailability with the IV AUC (area under the curve) underneath.

UNIT

188 mcg hr/mL = 188 (mg/L)hr, because mcg/mL and mg/L are the same concentration

One microgram per millilitre is one milligram per litre, so the two AUCs are already in the same unit and cancel. The doses are both 500 mg, so the dose ratio is 1.

ALGEBRA

$$F = \frac{188}{250} = 0.752$$

With equal doses F is the AUC ratio alone. The same answer comes from $F = \frac{Cl \times AUC_{po}}{D_{po}} = \frac{(2\ L/hr)(188\ (mg/L)hr)}{500\ mg} = 0.752$.

ROUND

F = 0.75

Her value, 75 per cent. Written as 0.75 or as 75 per cent; three quarters of the tablet dose becomes available for the body to use.

DOES THE NUMBER MAKE SENSE?

Equal doses, so F is just the area ratio: 188 is three-quarters of 250, so F is about 0.75. F must lie between 0 and 1.